Vinson Mervin

Computer Science Student | Databases, Data Analytics & Data Engineering
South Tangerang, Banten, Indonesia | +62 811-7712-609 | vinson.mervin2006@gmail.com
linkedin.com/in/vinson-mervin | github.com/vie-m

Summary

Computer Science student at BINUS University focused on databases. Designs relational schemas, builds ETL pipelines that turn raw data into clear answers, and builds full-stack web apps on PostgreSQL. Seeking an internship in data analytics or data engineering.

Education

BINUS University 2024 – Present
Bachelor of Computer Science, Focus: Databases *GPA: 3.41 / 4.00*

Projects

Olist E-commerce Data Warehouse | *PostgreSQL, Pentaho, SQL, Python* GitHub

- Built a **star schema** warehouse (112,650-row fact table, 5 dimensions) from 9 raw CSV files covering about 100K Brazilian e-commerce orders
- Wrote a **Pentaho ETL** job and an equivalent SQL + Python pipeline; both give identical row counts, revenue, freight and late-order totals
- Added data quality checks, 12 business SQL queries, RFM segmentation and 3 Matplotlib dashboards
- Found that late orders average **2.57 stars vs. 4.29** for on-time ones; only **3.0%** of 94,983 customers reordered

VieMeal, Meal Planner & Nutrition App | *React, Node.js, Express, PostgreSQL* Live Demo | GitHub

- Built a full-stack PWA for calorie and macro targets, food diary, progress charts and meal plans; deployed on Vercel + Neon
- Designed a **19-table PostgreSQL schema in 3NF**, queried with raw parameterized SQL (no ORM)
- Imported **7,793 USDA foods** plus 84 Indonesian dishes; added **typo-tolerant search** using `pg_trgm` with a GIN index, filtered by diet and allergies
- Wrote a rule-based meal planner that keeps each day within **±10% of the calorie target**; the optional server-side AI assistant (Gemini) never supplies nutrition numbers, all come from the database

Tiket-IN, Event Ticketing | *React, Node.js, Express, PostgreSQL* Live Demo | GitHub

- Built a full-stack ticketing app with accounts, event browsing, seat map and admin sales report on a **7-table schema in 3NF** (raw parameterized SQL, no ORM); deployed on Vercel + Neon
- Prevented double booking with a **transaction, row locking (SELECT ... FOR UPDATE)** and a UNIQUE constraint; in a test of 10 simultaneous bookings for one seat, 1 succeeded and 9 were rejected

YouTube Trend Analysis | *Python, YouTube Data API, SQLite, Streamlit* Live Demo | GitHub

- Collected and analyzed **3,000 videos across 15 tech-review channels** with YouTube Data API, Python and SQLite; cached raw responses and built an idempotent loading pipeline
- Created a **channel/format/age-group normalized performance index**, SQL analysis and Streamlit dashboard; found **306 breakout videos** at least 3x their comparison-group median

Skills

Languages: Python, SQL, JavaScript, Java, C, R, HTML/CSS
Data & Databases: PostgreSQL, star schema / dimensional modelling, 3NF design, Pentaho Data Integration (ETL), data quality checks, query tuning (EXPLAIN ANALYZE)
Libraries: pandas, Matplotlib
Web: React, Vite, Tailwind CSS, Node.js, Express, REST APIs, JWT authentication
Tools: Git, GitHub, Jupyter, Vercel, Neon
Soft Skills: Teamwork, analytical thinking, problem-solving, attention to detail, discipline